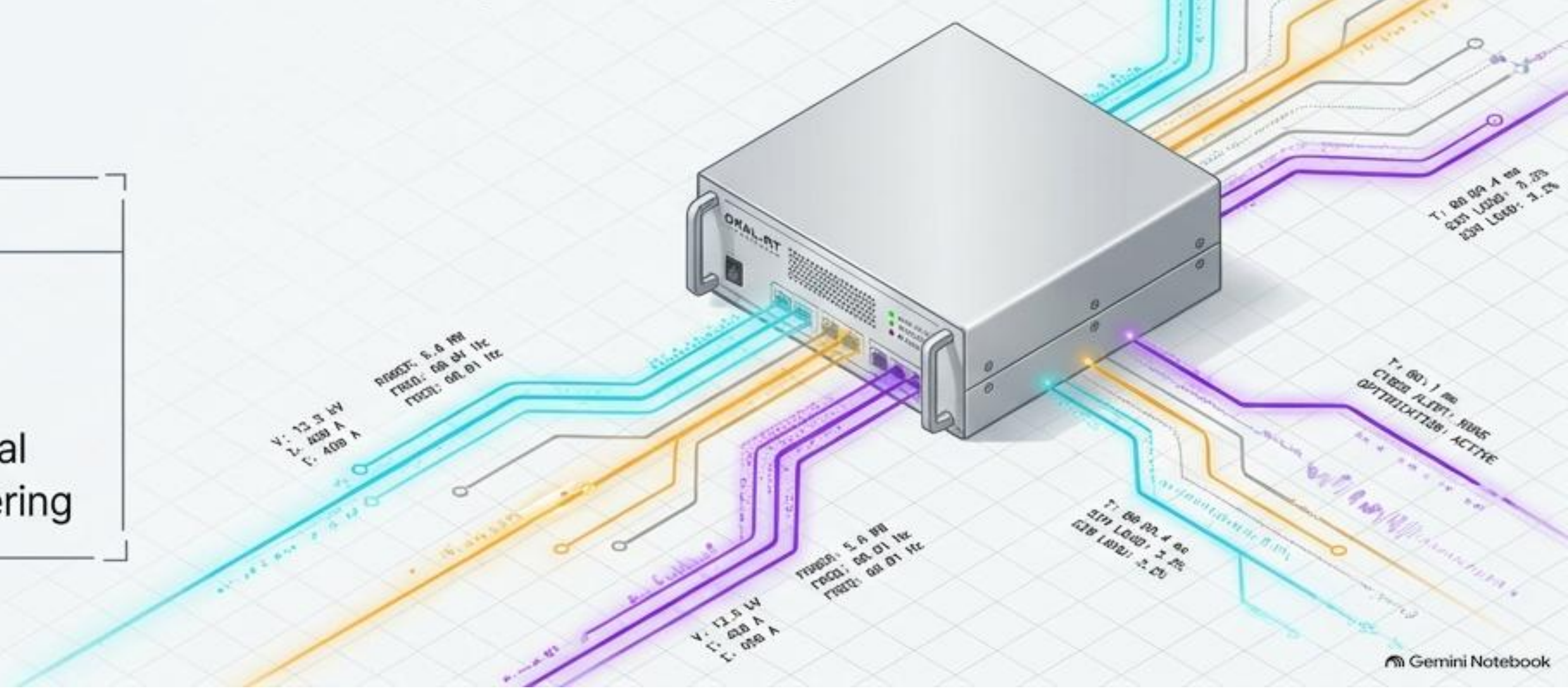
Cognitive Adaptive Power System Management (CAPSM)

A Brain-Inspired AI Framework Validated Through Sub-10ms Hardware-in-the-Loop Testing



MODULAR TELEMETRY

Author: Mahmoud Kiasari

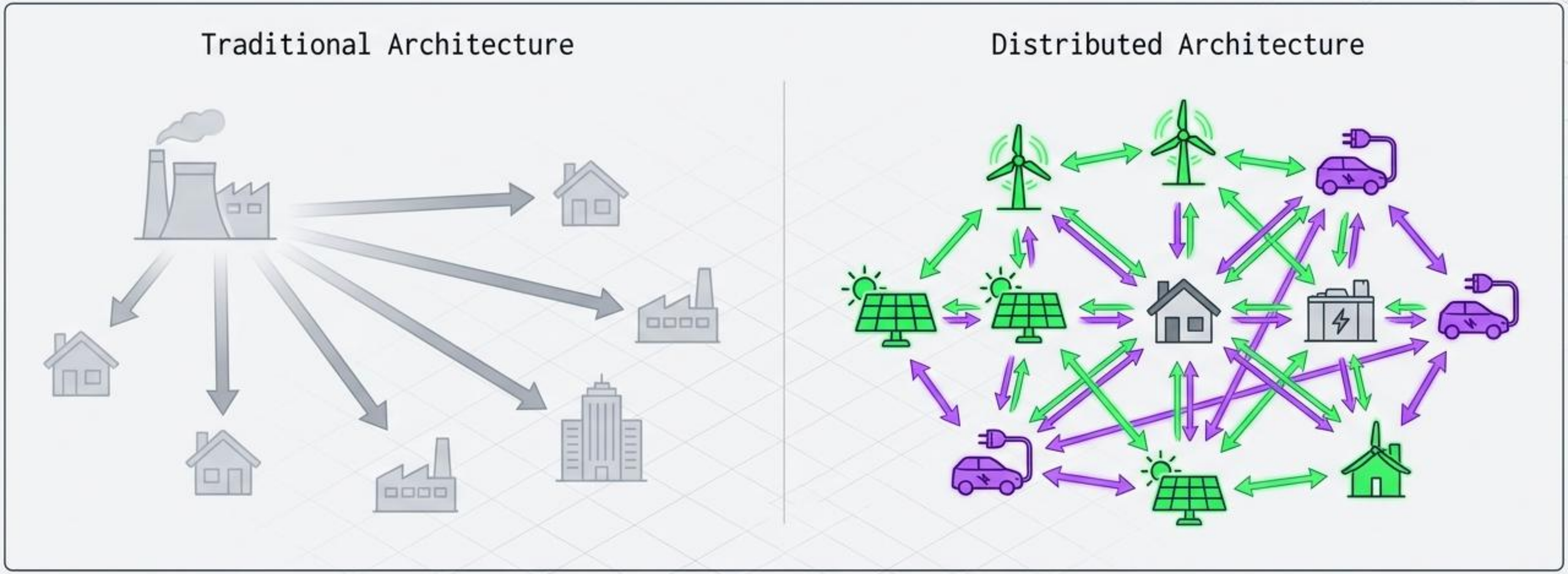
Institution: Dalhousie University

Department of Electrical and Computer Engineering

The Intermittency and Complexity Crisis

Global Renewable Capacity: +60% (Reaching 4,800 GW)

DER Growth: 132.4 GW to 528.4 GW (16.7% CAGR)

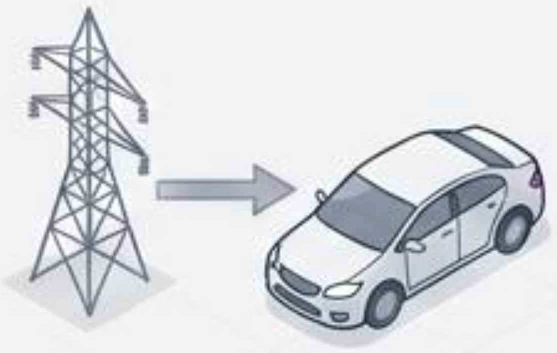


Reduced System Inertia
(Sync generators replaced by inverters)

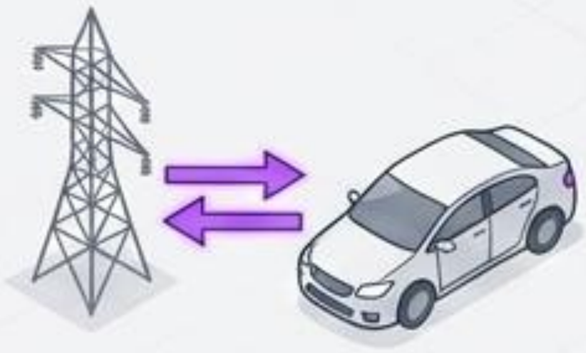
Bidirectional Power Flows
(Distribution networks behaving like transmission lines)

Cybersecurity Vulnerabilities
(Digitalisation of critical infrastructure)

Vehicle-to-Grid (V2G) Architecture Concept



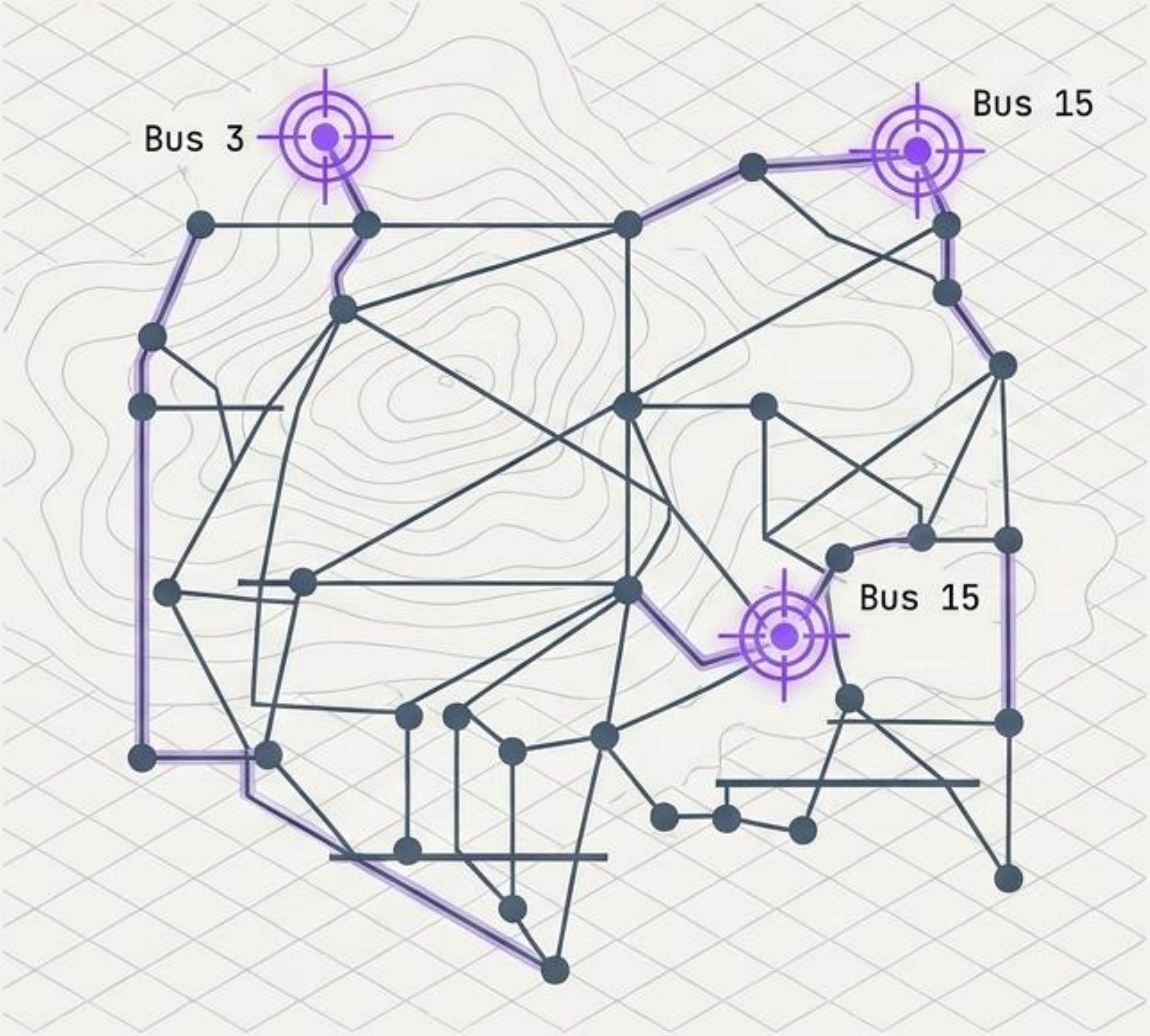
G2V: One-Way Flow

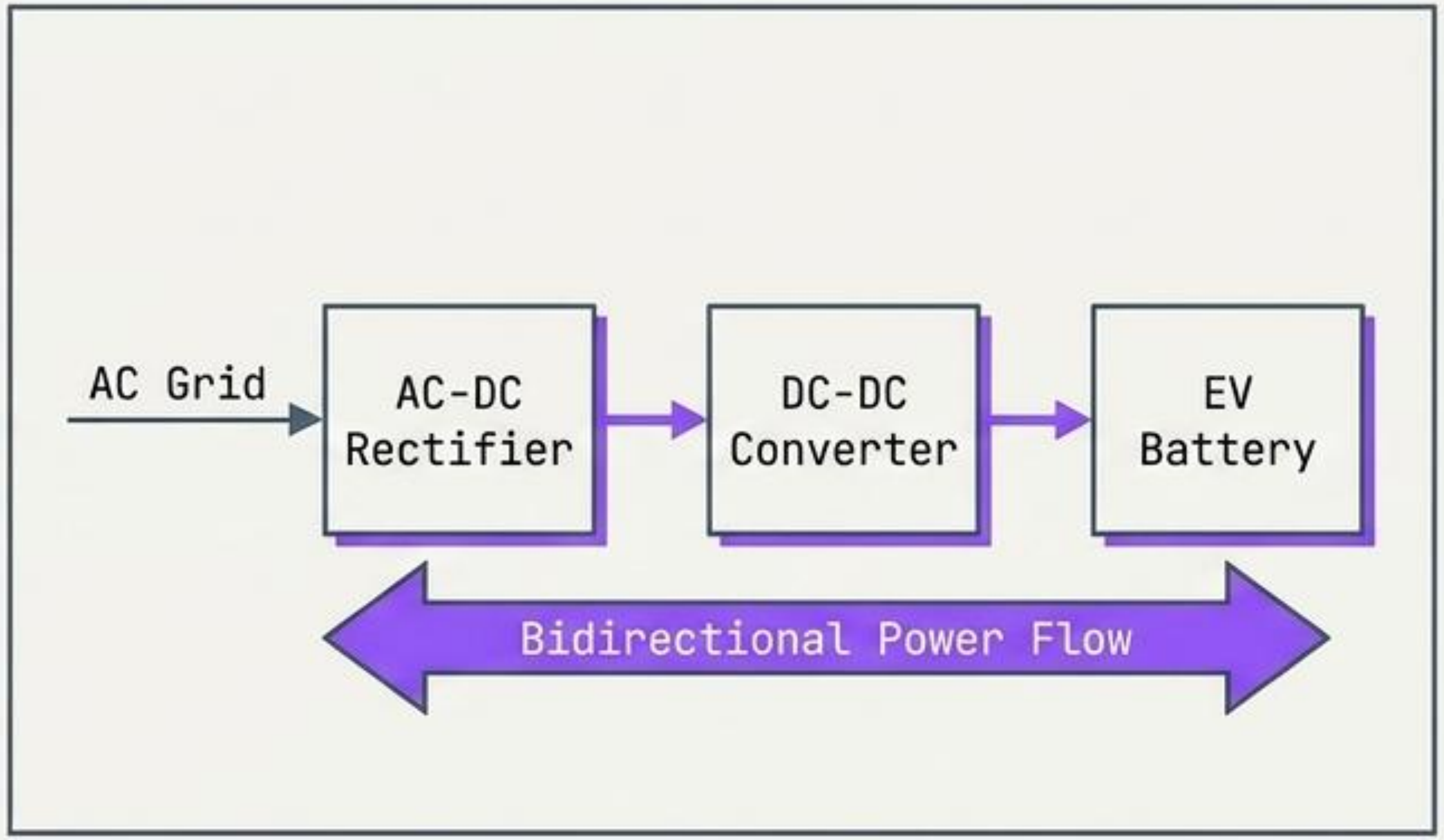


V2G: Two-Way Flow

	Grid-to-Vehicle (G2V)	Vehicle-to-Grid (V2G)
Power Flow	Unidirectional	Bidirectional
Consumer Role	Passive consumer	Active Distributed Energy Resource (DER)
Grid Impact	Acts as grid burden	Provides frequency regulation & storage
Hardware Requirements	Standard rectifiers	Complex bidirectional power electronics and Phase-Locked Loop (PLL) synchronisation

EV Charger Modeling and Placement

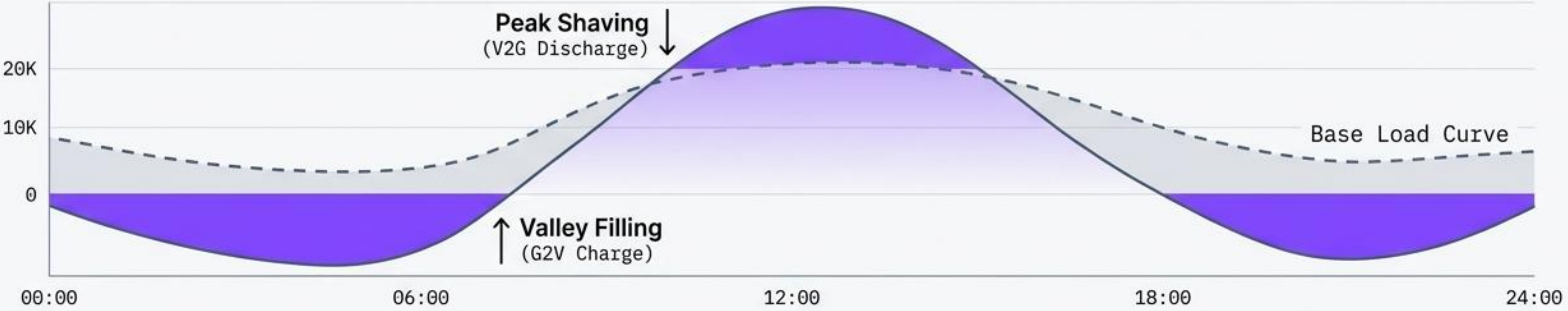




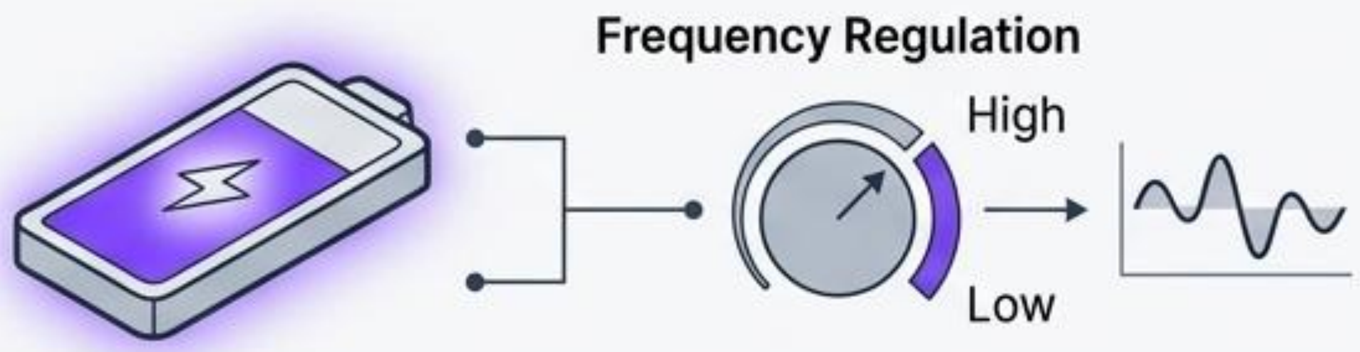
CAPSM EV Fleet Parameters:
- 3 EV fleets modeled
- 19.2 kW power capability per unit
- 40 kWh capacity per unit

Smart Charging Strategies & Grid Support

Dynamic Daily Load Curve

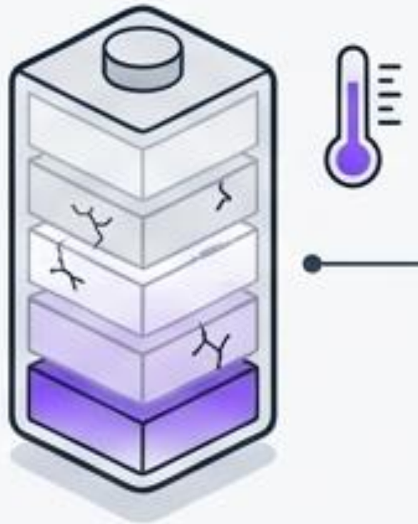


Frequency Regulation



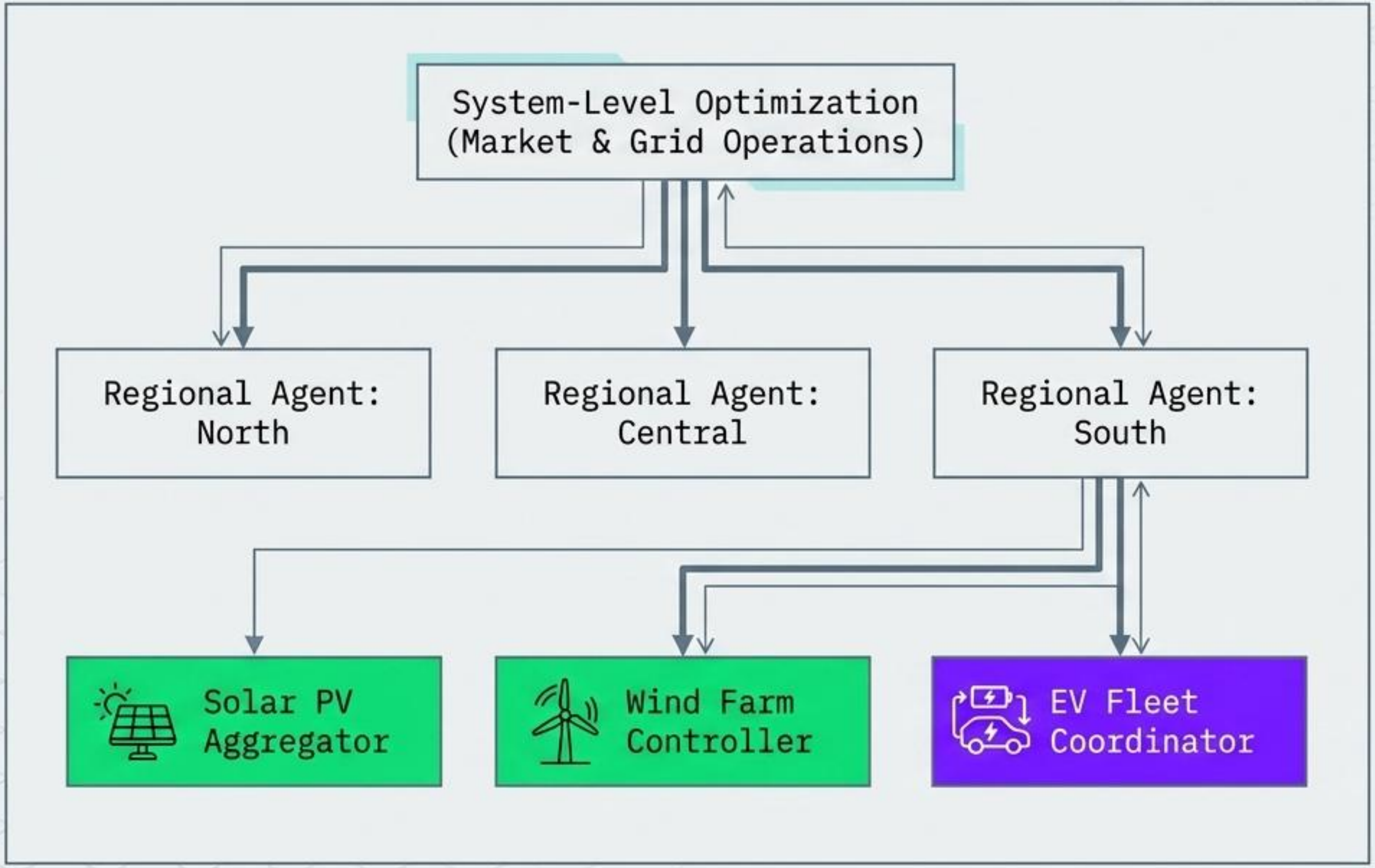
Primary & Secondary Regulation: Dynamic power dispatch based on droop and AGC signals.

Degradation Management



Degradation Management: Balancing grid support against battery cycle limits to ensure daily commute capacity.

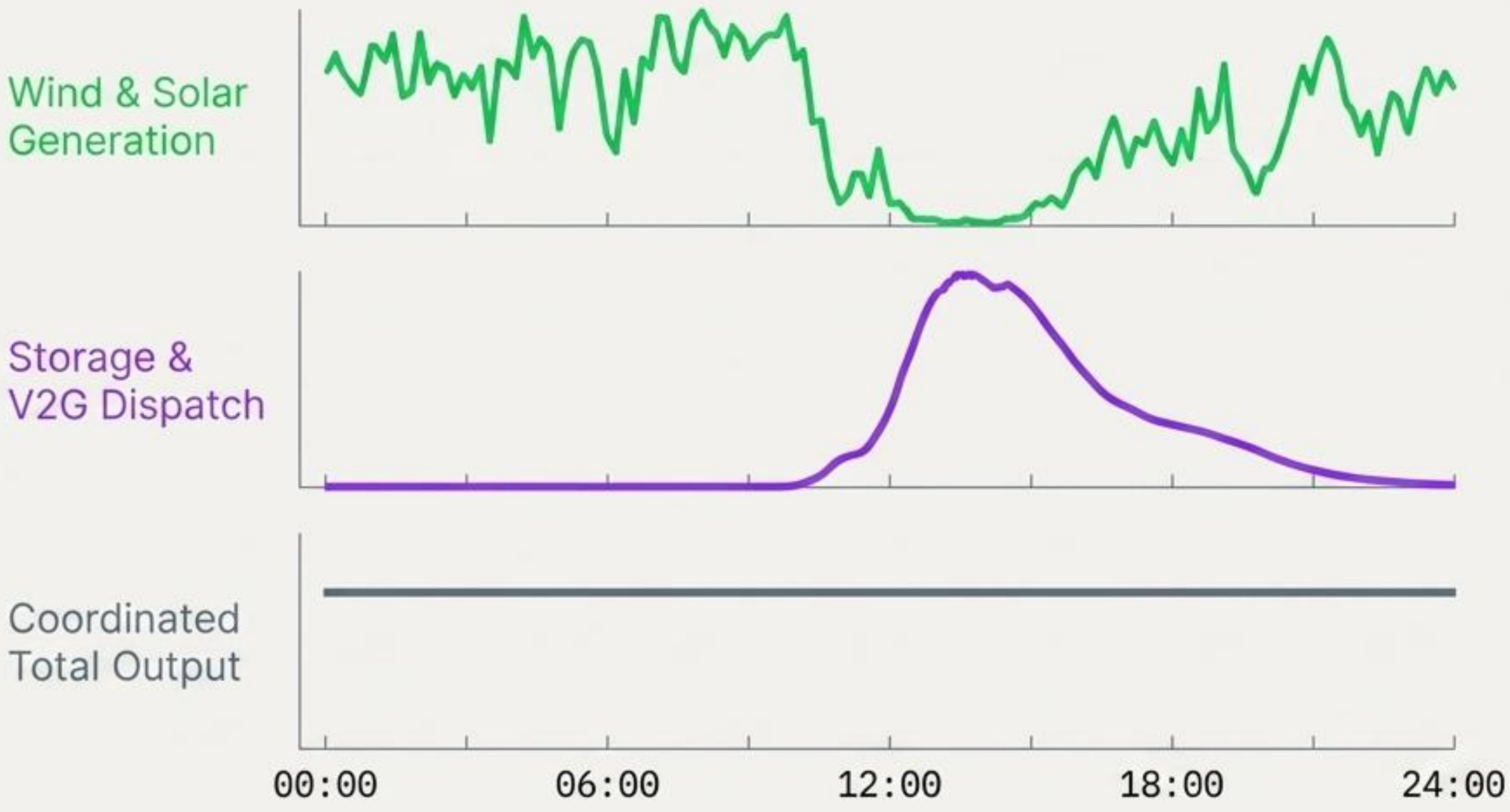
DER Management & Coordination Framework



- **Hierarchical Architecture:** Distributes computational load and prevents central bottlenecks.
- **Multi-Agent Coordination:** Employs ADMM consensus algorithms to synchronize independent DERs.
- **Temporal Coordination:** Aligns fast-acting inverters with slow-moving mechanical generation.

Gemini Notebook

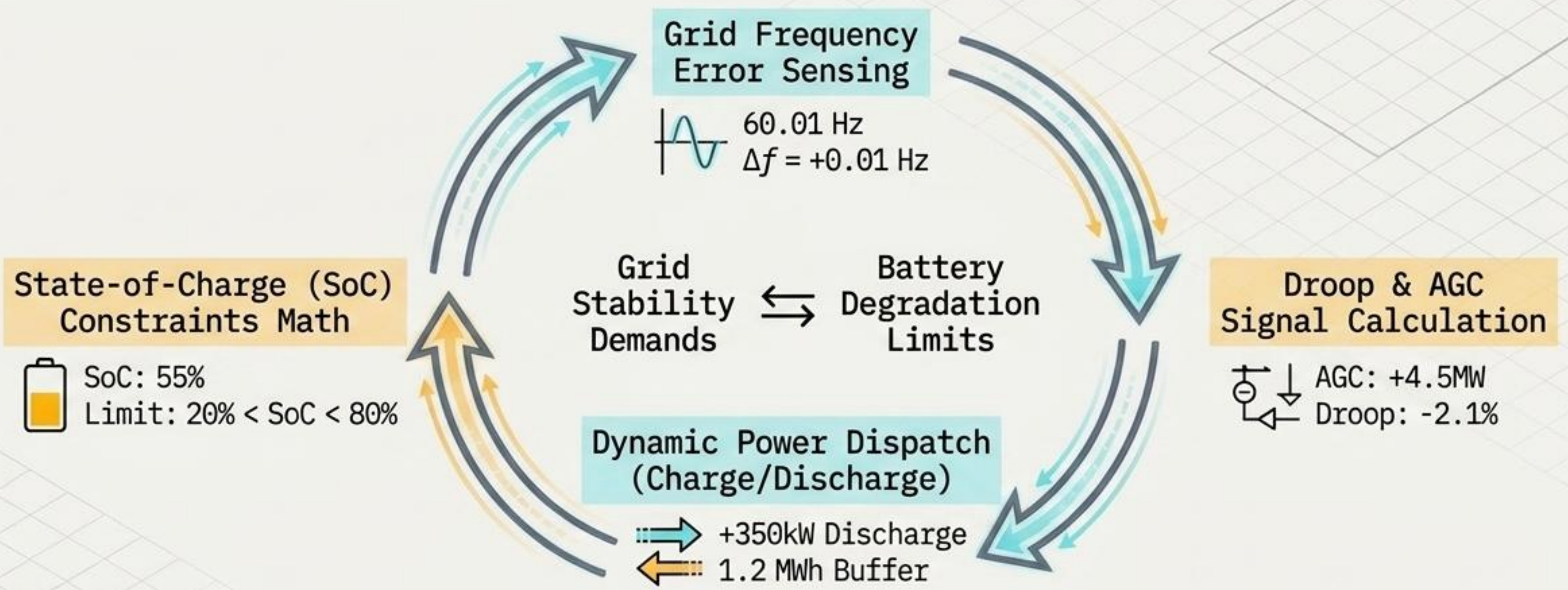
Renewable Energy Coordination



Intermittency
Smoothing Profile

1. Senses weather-dependent generation dips via PMU data.
2. Temporally coordinates battery storage buffers.
3. Results in a flat, predictable grid injection.

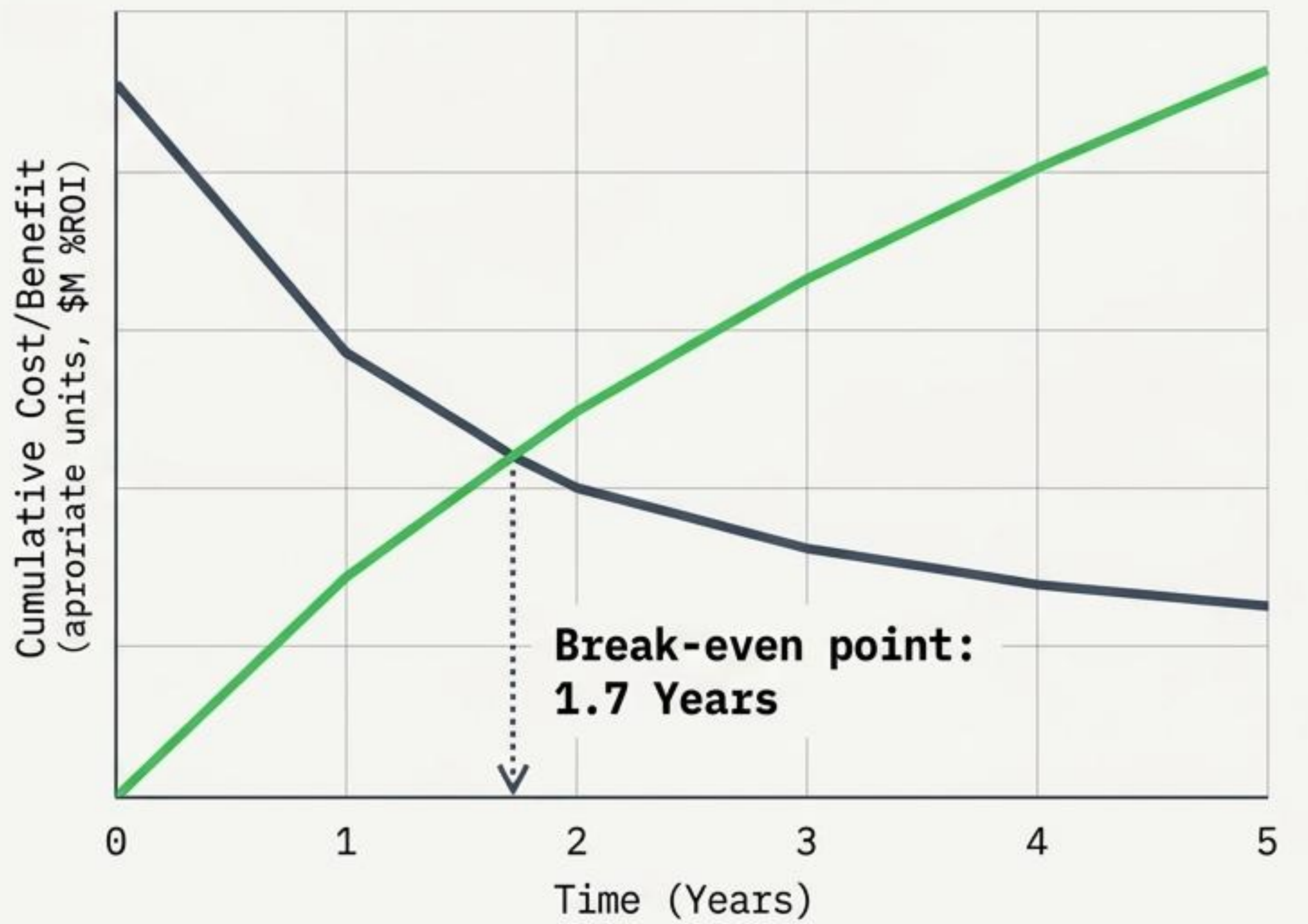
Battery Coordination & SoC Management



Maintains optimal battery levels for consumer EV driving schedules.

Constrains V2G cycling to prevent accelerated capacity fade and semi-empirical degradation.

Economic Impact & V2G Simulation Results



12.4%

Annual operational cost reduction
($51.5M/year on IEEE 118-bus)

42.5%

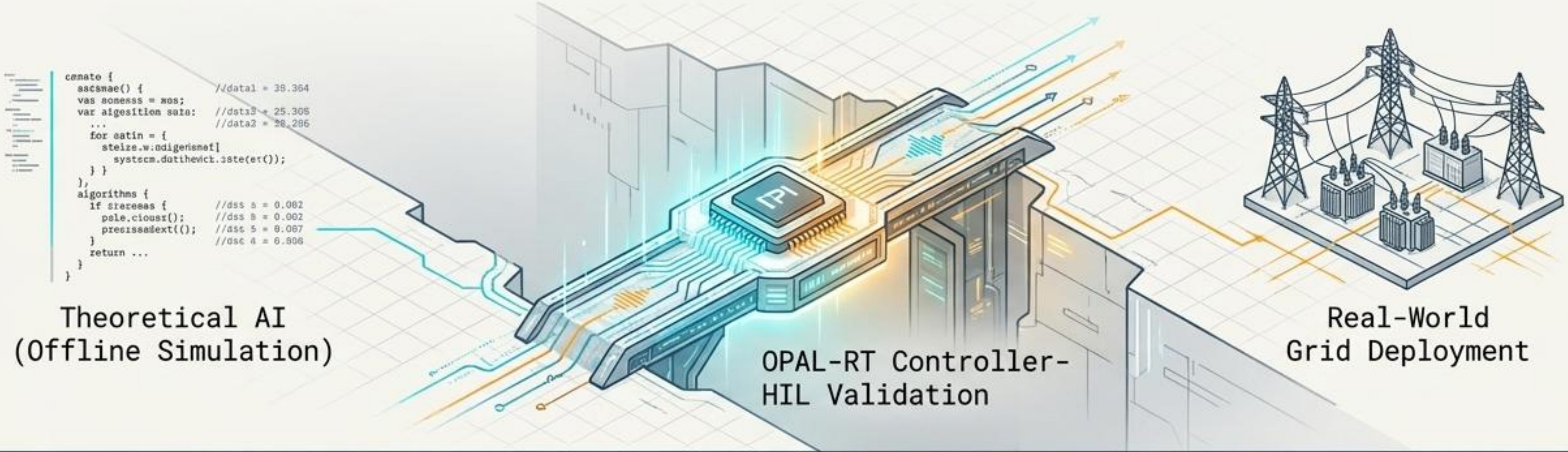
Reduction in renewable energy
curtailment

28.0%

Increase in Distributed Energy
Resource (DER) utilisation efficiency

Integrating wind and solar intermittency with V2G dispatch yields an **83.7%** **Internal Rate of Return**.

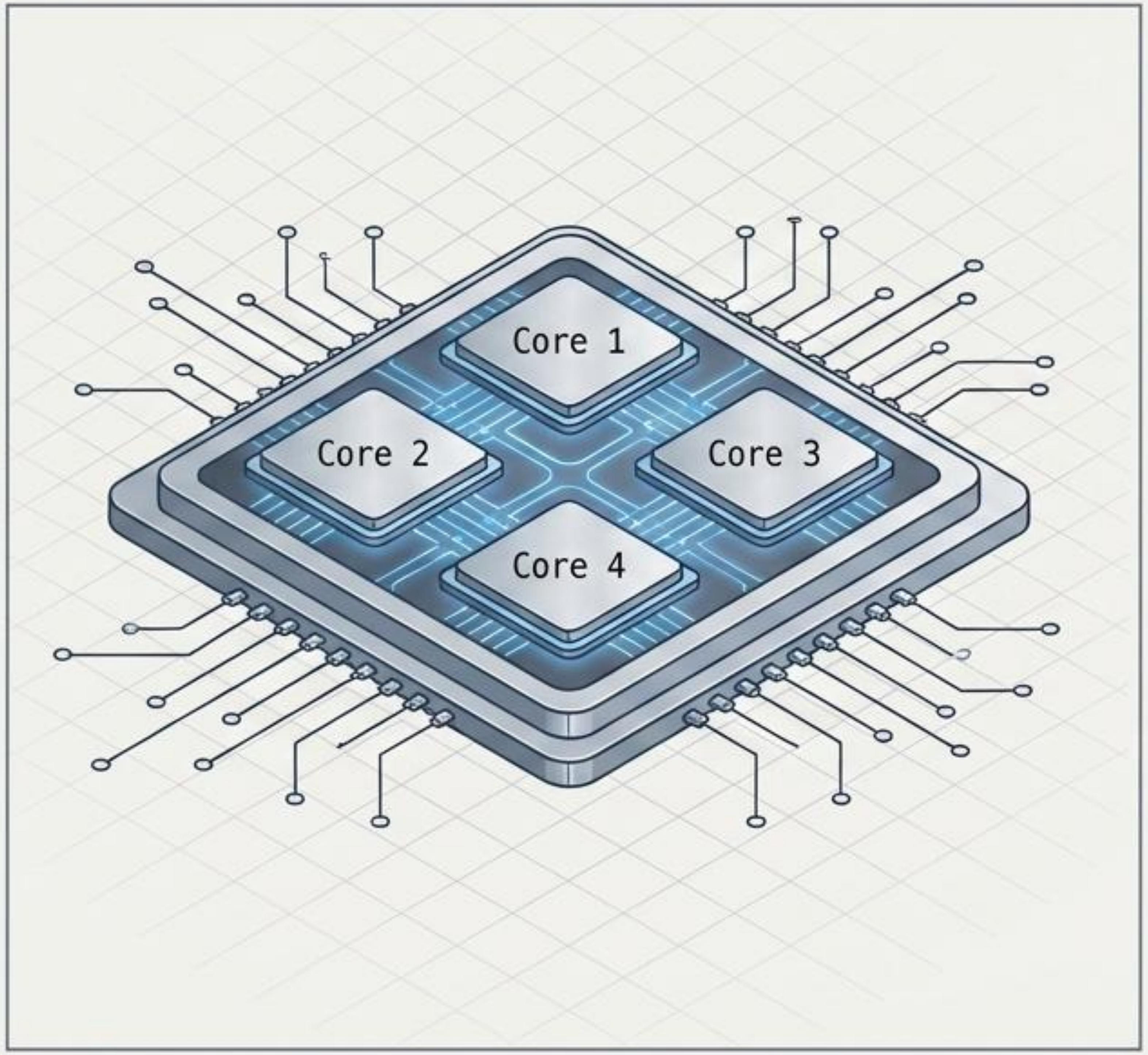
The Validation Gap: Moving AI to Physical Hardware

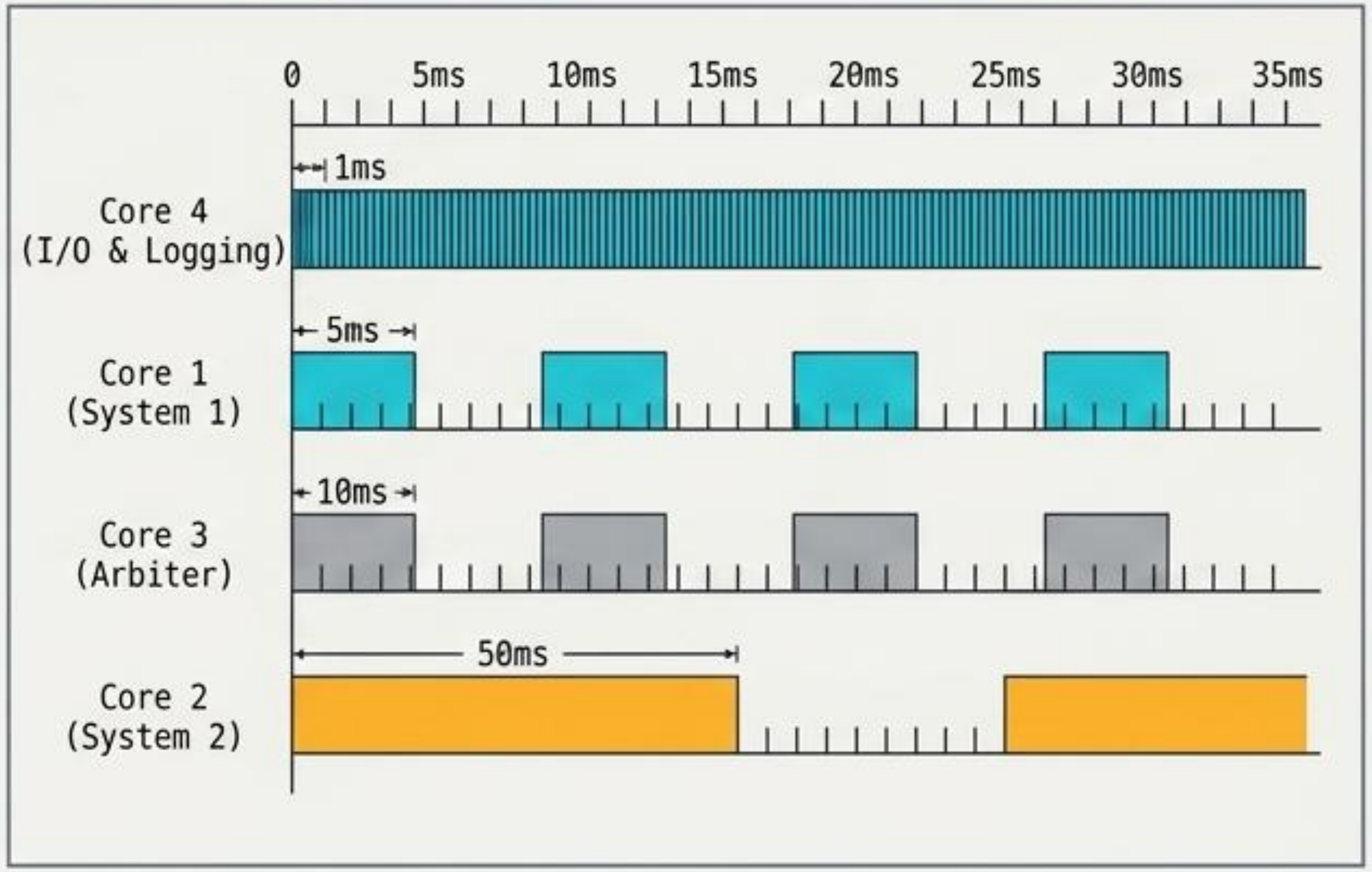


Constraint	Offline Simulation	Hardware-in-the-Loop (HIL)
Time Management	Pauses for infinite compute	Strict real-time milliseconds
Latency	Idealised zero delay	Actual physical comms delay
Processing Limits	Infinite cloud computing	Hardware-bound (4-Core)
Validity Level	Theoretical Math Proof	TRL-6 Field Ready

Theoretical AI models often fail in the field due to communication latency and real-time execution limits. HIL validation is the essential bridge.

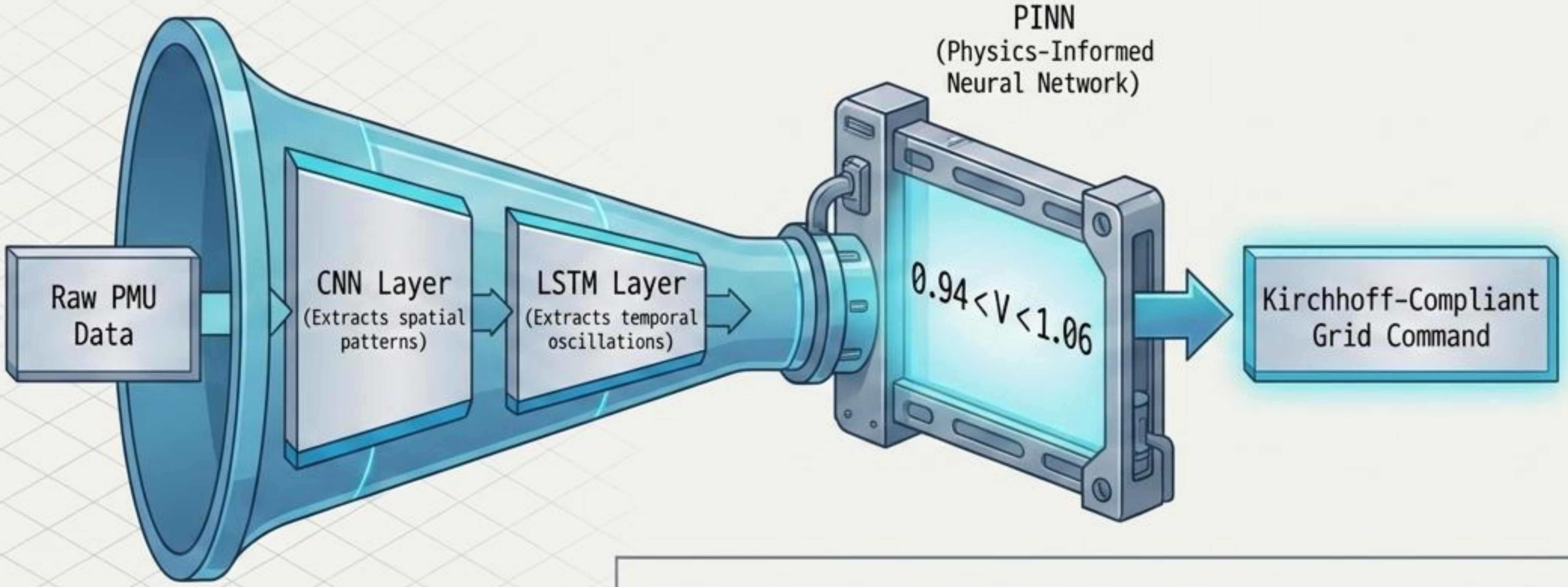
4-Core OPAL-RT Architecture & Task Partitioning





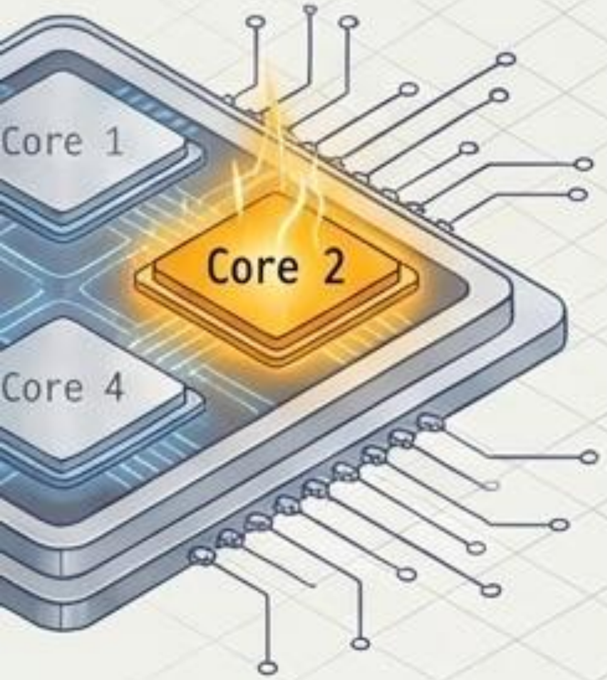
Multi-timescale AI operation requires strict rate-monotonic scheduling. High-frequency I/O fires rapidly while cognitive planning runs on longer, overlapping cycles.

Core 1: System 1 CNN-LSTM Reflex

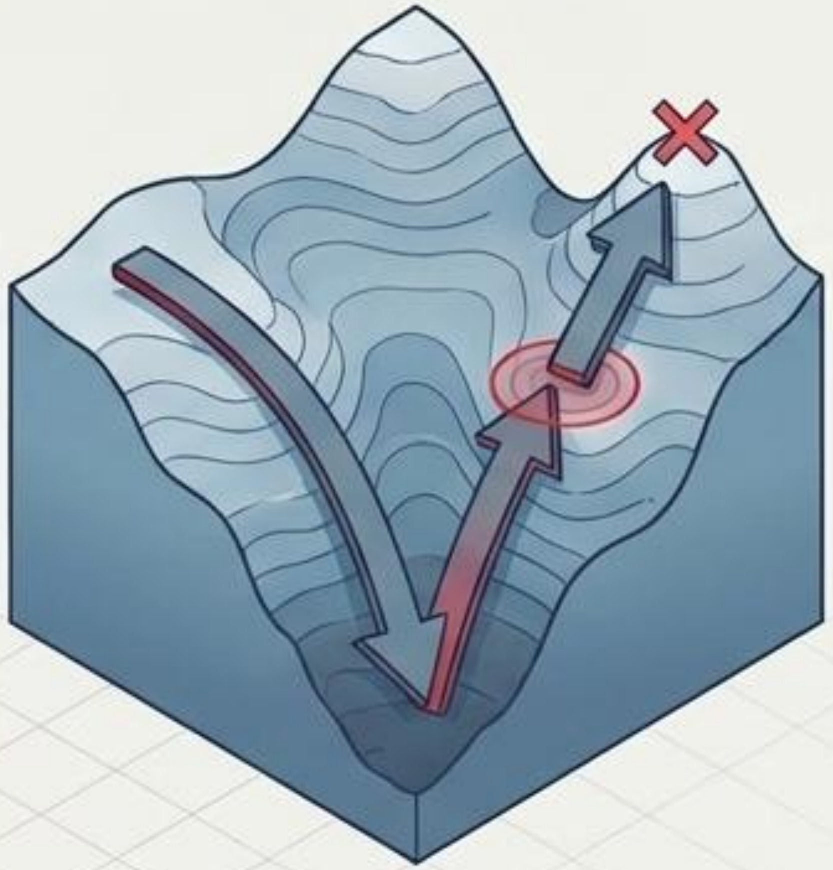


Assignment: North Region (Buses 1-19)
Constraint: ⏱ 5ms timestep (Highest Priority)
Function: Sub-millisecond inference for immediate grid protection and fault detection.

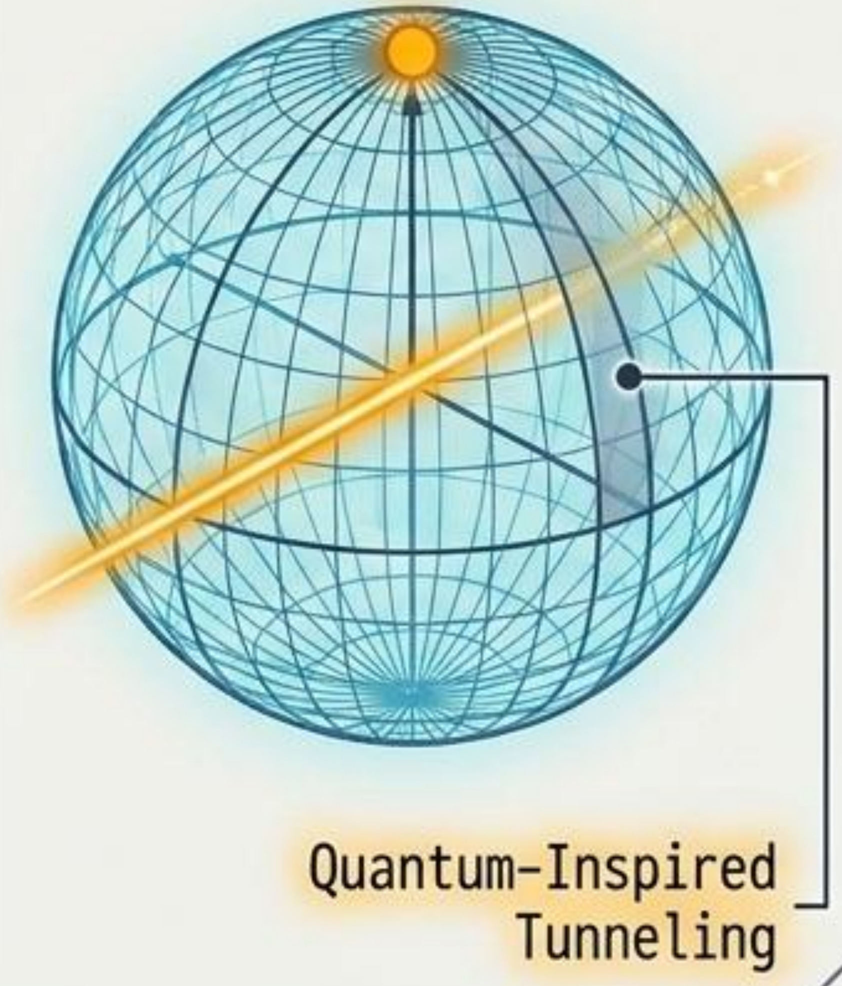
Core 2: System 2 QIRL Optimisation



Standard Algorithm
(Trapped in Local Optimum)

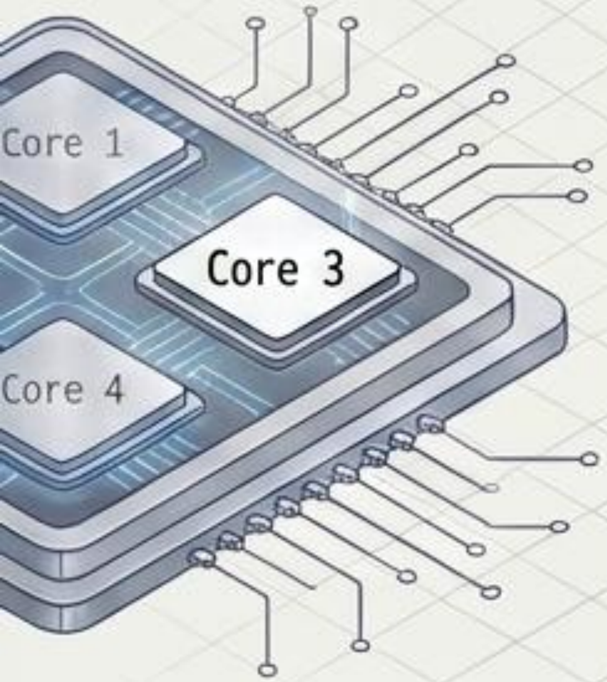


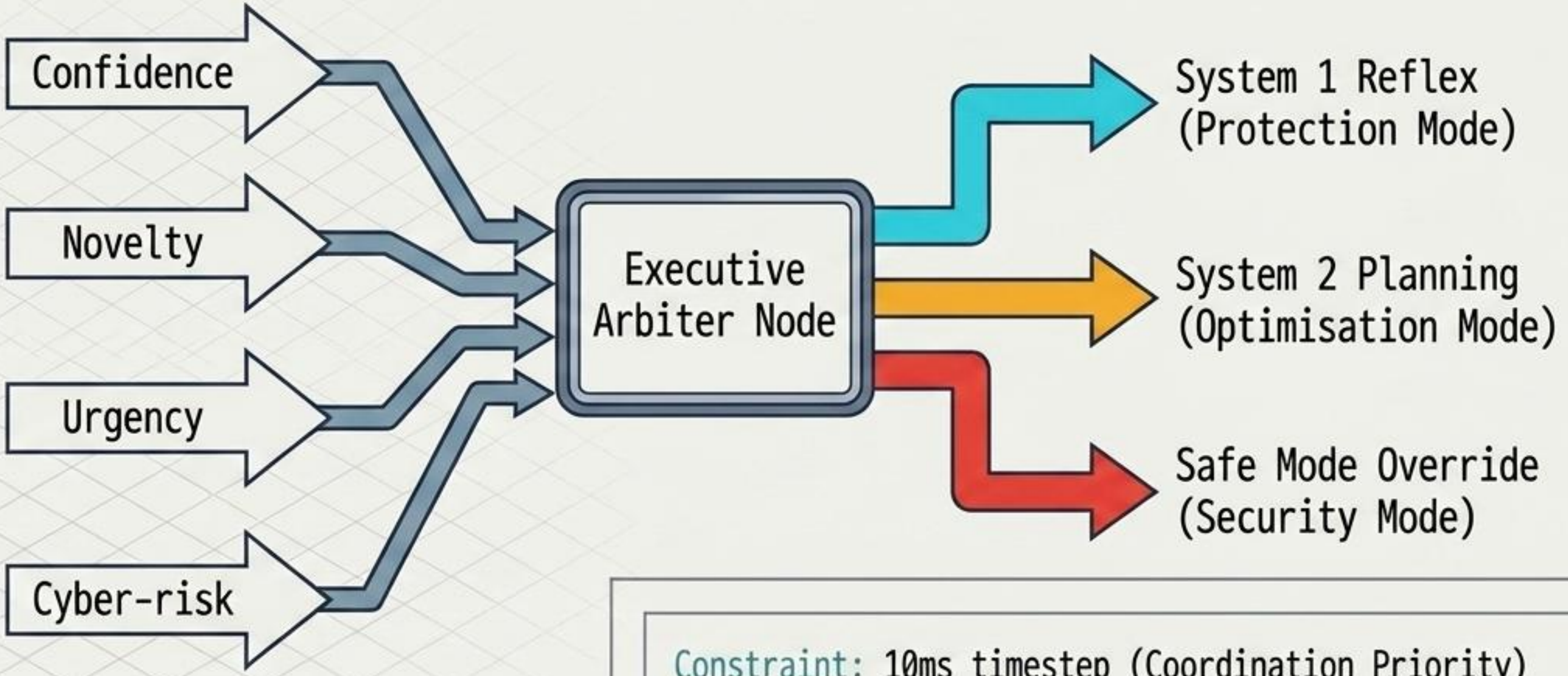
Quantum-Inspired Tunneling
(Finds Global Optimum)



Assignment: South Region (Buses 20-39)
Constraint: 50ms timestep (Medium Priority)
Performance: Quantum-Inspired Reinforcement Learning (QIRL) achieves **58% faster** convergence than classical Deep Q-Networks (DQN) for economic dispatch and scheduling.

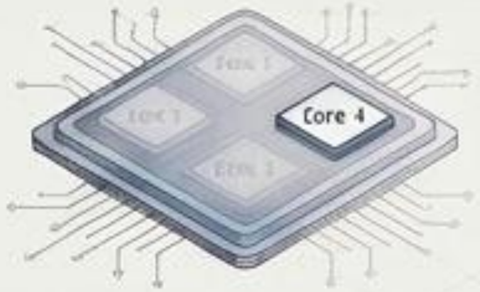
Core 3: Metacognitive Arbitration



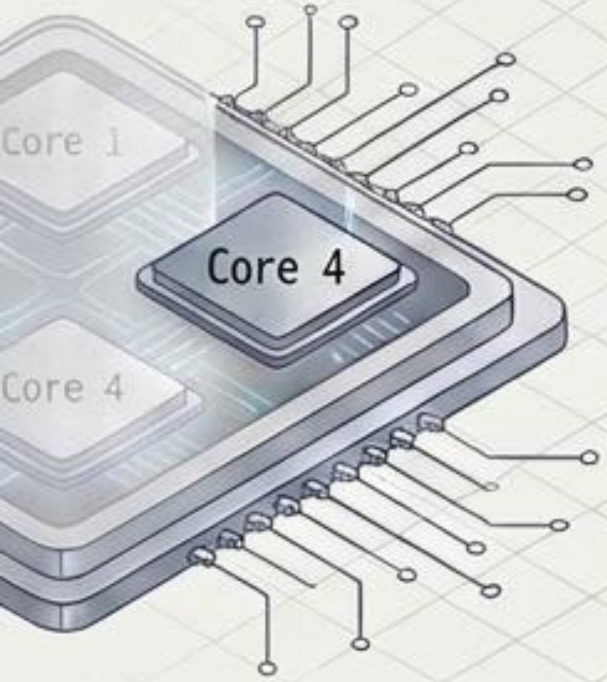


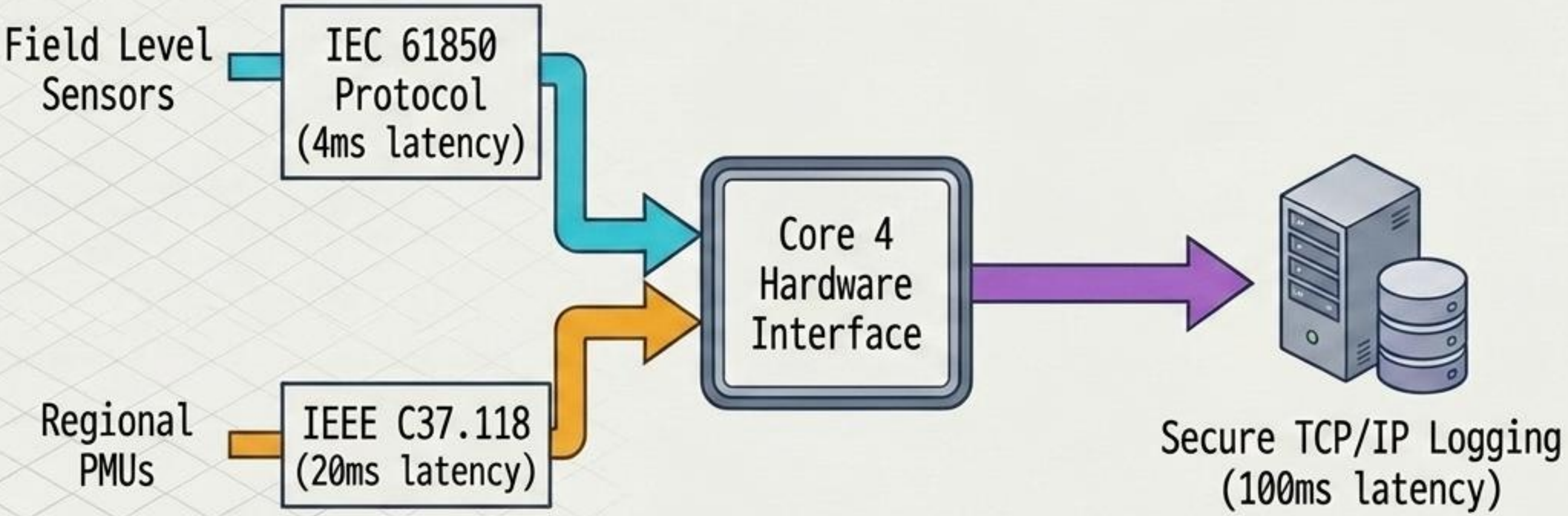
Constraint: 10ms timestep (Coordination Priority)

Function: Manages the tension between fast protection and slow optimisation, instantly reverting to safe physical setpoints if cyber-anomalies are detected.



Core 4: I/O and Logging Interface





Constraint: 1ms timestep (Lowest Computational Priority)
Function: Dedicated exclusively to physical hardware synchronization, data logging, and HIL I/O interfaces to prevent network traffic from interrupting AI inference on Cores 1-3.

RT-LAB Configuration & The 5-Scenario Gauntlet

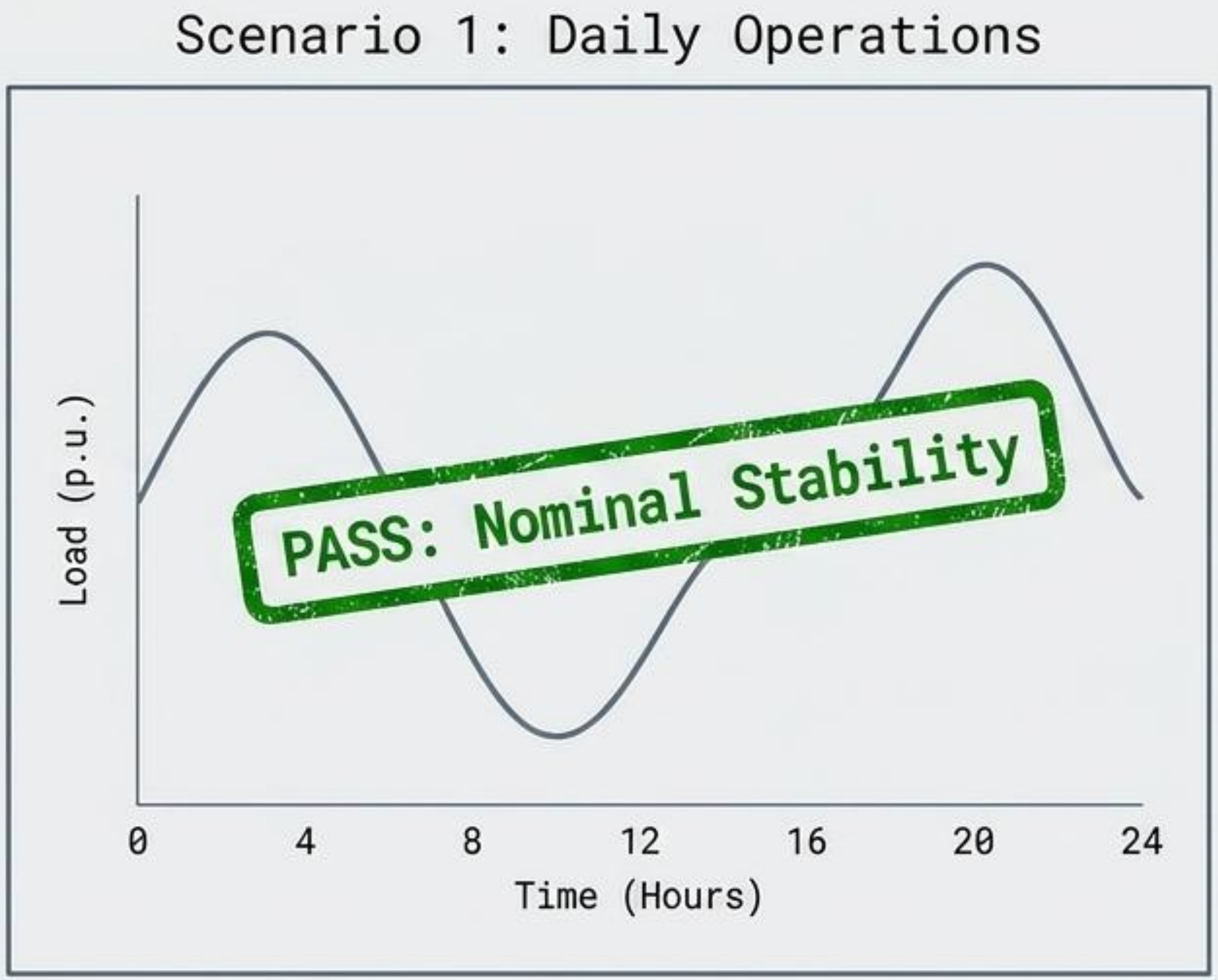
OPAL-RT RT-LAB Configuration

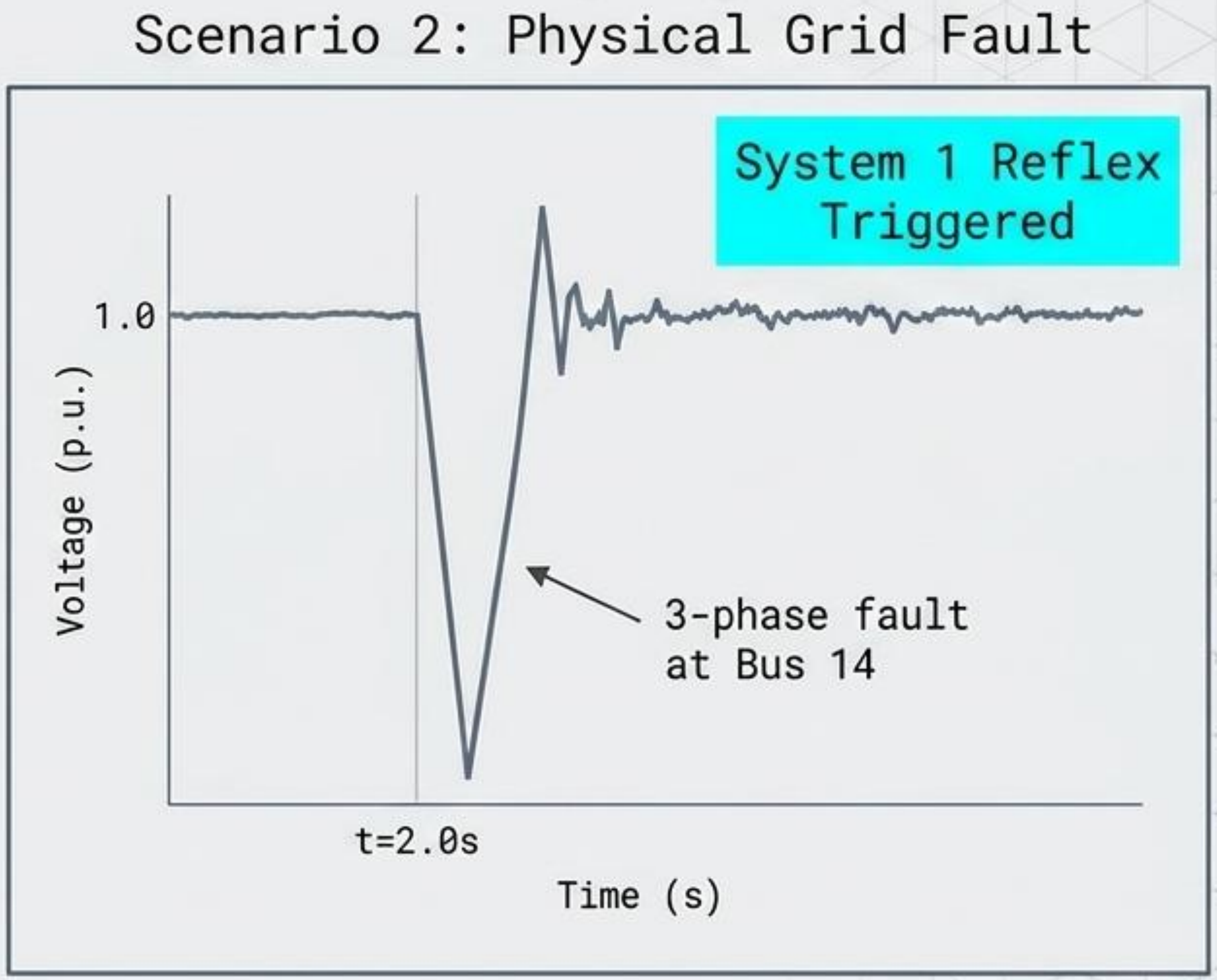
ARTEMIS Solver locked at strict 50µs Electromagnetic Transient (EMT) timestep.

Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5
Normal Operation	Fault Response	Intermittency	Cyber Attack	EV Fleet
(Baseline tracking)	(3-phase short circuit)	(Wind generation drop)	(False Data Injection)	(V2G Coordination)

Note: The following telemetry results are recorded live from physical OPAL-RT hardware, not simulated offline outputs.

Telemetry: Normal Operation & Extreme Faults





Fault Detection Metrics: Achieved with 95.8% accuracy and 1.73% localisation error, preventing cascading grid failure.

Telemetry: Intermittency & Cyber-Physical Resilience

Scenario 3: Renewable Drop

WIND GENERATION (MW)

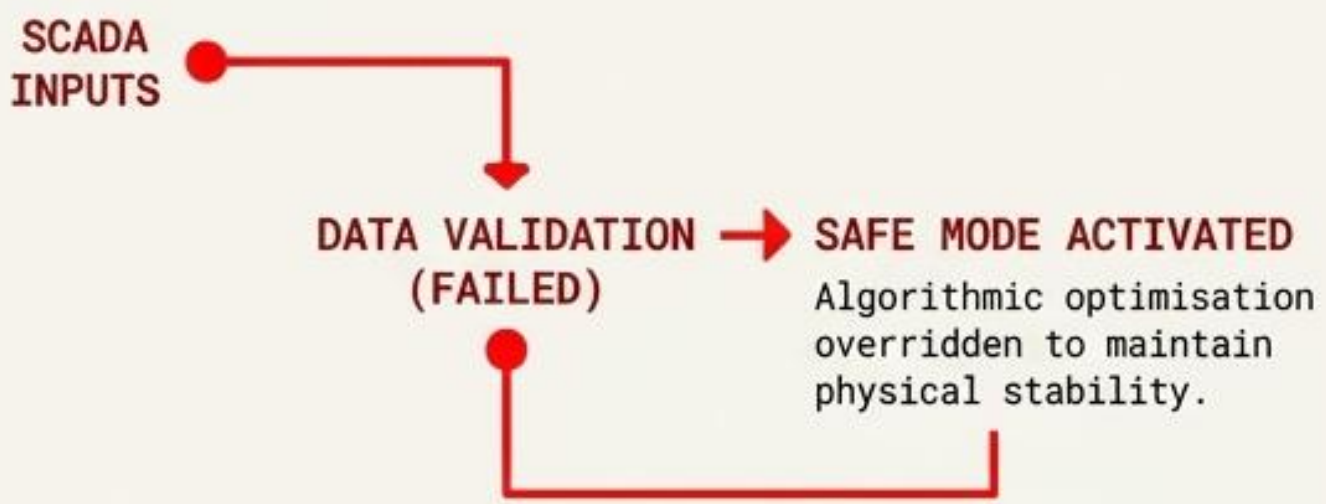
SYSTEM 2 BATTERY DISPATCH (MW)

SYSTEM FREQUENCY (Hz) — STEADY

60.00 Hz

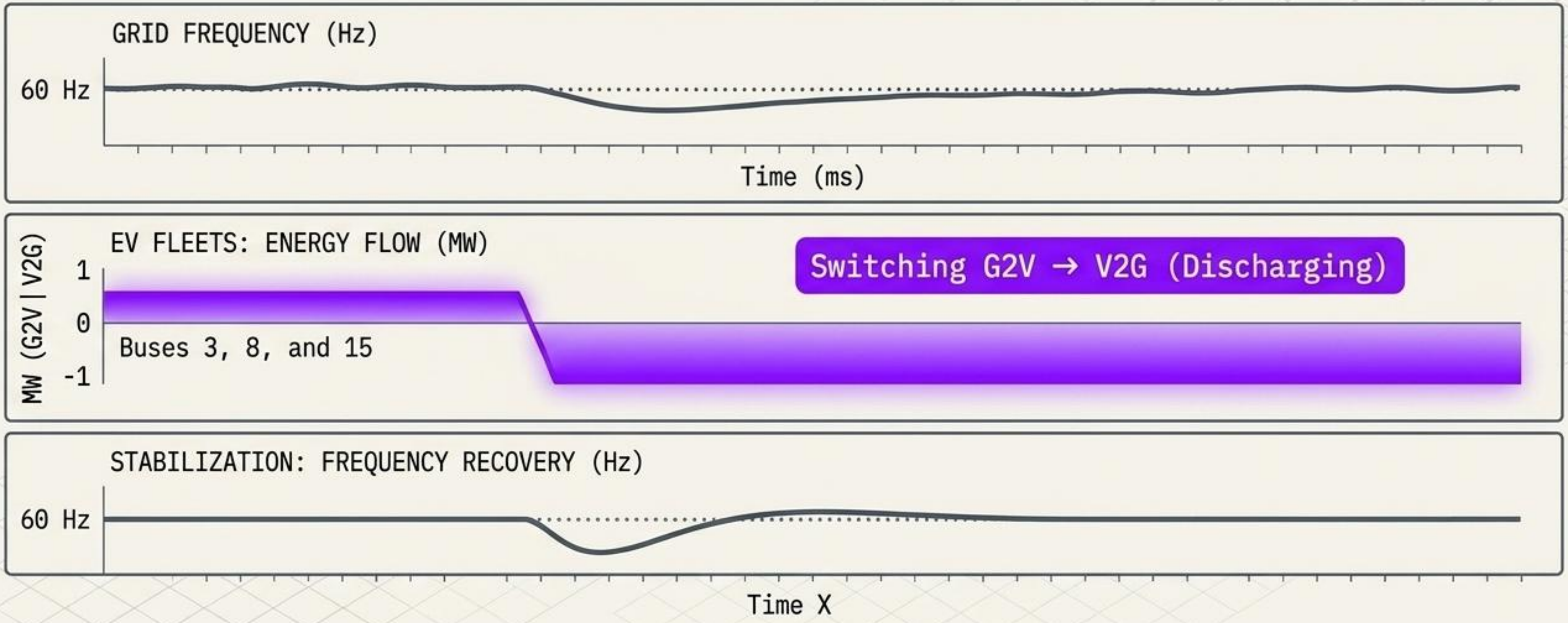
Scenario 4: Cyber Attack (FDI)

CRITICAL ALERT: False Data Injection detected at Buses 5 and 10



The Metacognitive Arbiter successfully manages both environmental variability and malicious external data poisoning.

Scenario 5: Real-Time EV V2G Coordination



Hardware Validation Confirmed: Bidirectional EV chargers successfully synchronized via Phase-Locked Loops and executed smart charging strategies within strict 50μs solver constraints.

The Reality of CAPSM: Hardware Validated, Economically Proven

Real-Time Response Confirmed on physical OPAL-RT silicon.

The 12.4% economic savings and 37% stability enhancements are no longer theoretical. The brain-inspired dual-process AI successfully controlled a 39-bus power system in sub-10ms conditions and is ready for industry integration.